

YOUNGJIN HONG

(763)388-7740 | hong0745@umn.edu | [LinkedIn](#) | [Google Scholar](#) | [Website](#)

RESEARCH INTERESTS

Vision-Language-Action Models, Multimodal Robot Learning, Generative Policies, Contact-Rich Manipulation

EDUCATION

University of Minnesota (UMN)

Ph.D. Electrical Engineering

Minneapolis, MN, USA

Sep. 2024 – Present

Sungkyunkwan University (SKKU)

M.S. / B.S. in Mechanical Engineering

Suwon, Korea

Mar. 2016 – Feb. 2024

RESEARCH & INDUSTRY EXPERIENCE

Research Intern [\[Website\]](#)

Mitsubishi Electric Research Laboratories (MERL), Optimization & Robotics Team

Cambridge, MA

May. 2026 – Sep. 2026

- Developed learning-based control methods for robust contact-rich robotic disassembly under pose uncertainty.
- Integrated learned manipulation policies with compliant control on an industrial robot platform.
- Led real-robot evaluation across multiple disassembly tasks, resulting in a first-author ICRA 2027 submission.

Research Assistant [\[Website\]](#)

Choice Robotics Lab, University of Minnesota

Minneapolis, MN

Sep. 2024 – Present

- Conduct research on Vision-Language-Action models and computer vision for real-world applications.
- Develop learning-based robotic systems using simulation and physical platforms including MuJoCo and ALOHA2.
- Train multimodal models and conduct large-scale experiments on GPU clusters using PyTorch and SLURM.

Robot Research Engineer [\[Website\]](#)

Hanwha Aerospace, Manned-Unmanned Teaming (MUM-T) Research Center

Seoul, Korea

Jan. 2024 – Jul. 2024

- Built Isaac Sim environments and supported field testing for autonomous unmanned ground vehicle research.

SELECTED PROJECTS

VLM-Guided Person Re-Identification and Tracking (Collaboration with ETRI)

Sep. 2025 – Present

- Developed a language-guided person tracking pipeline integrating **YOLOv11-L**, **ByteTrack**, and **LLaVA-NeXT** for identifying and re-identifying a described target in continuous video.
- Designed an **adaptive VLM triggering mechanism** that invokes multimodal reasoning only during initialization, target loss, low-confidence tracking, or periodic verification.

Language-Action Cycles for Learning Manipulation Tasks

Jun. 2025 – Oct. 2025

- Proposed a **bidirectional language-action cycle** that jointly learns action-to-language and language-to-action mappings for scalable behavior acquisition.
- Designed a semantic consistency verifier for filtering self-generated samples, enabling reliable self-improvement.

SELECTED PUBLICATIONS / PRESENTATIONS

Youngjin Hong, Tian Xie, Mingen Li, et al. "NUDGE: Efficient Structured Action Correction for Vision-Language-Action Models under Latency," *Submitted to ICRA 2027*. [\[Website\]](#)

Youngjin Hong et al. First-author manuscript on learning-based compliant robotic disassembly under pose uncertainty. *Submitted to ICRA 2027*. Research conducted at MERL.

Hantao Ye, Ross Worobel, Zhuoli Xie, Mingen Li, Houjian Yu, **Youngjin Hong**, Changhyun Choi. "POINTCAST: One World Model for Rigid, Articulated, and Deformable Object Manipulation," *Submitted to ICRA 2027*. [\[Website\]](#)

Youngjin Hong*, Houjian Yu* (*equal contribution) et al. "LACY: A Vision-Language Model-based Language-Action Cycle for Self-Improving Robotic Manipulation", *ICRA 2026*. [\[Website\]](#) [\[Paper\]](#)

Mingen Li, Houjian Yu, Yixuan Huang, **Youngjin Hong** et al. "Hierarchical DLO Routing with Reinforcement Learning and In-Context Vision-language Models", *ICRA 2026*. **Best Paper Award on Robot Learning Finalist** [\[Website\]](#) [\[Paper\]](#)

SKILLS

Programming: Python, C++, MATLAB | **ML:** PyTorch, Hugging Face, Vision-Language Models (VLM), Vision-Language-Action Models (VLA), Diffusion/Flow Matching | **Robotics:** ROS, MuJoCo, Isaac Gym/Sim, ALOHA2, Franka, MELFA | **Infra:** Linux, Git, Docker, SLURM, GPU clusters